

HSC Investigating Science: Modules 5-8

Comprehensive Study Notes

Year 12 Course - Scientific Investigations and Critical Analysis

Synthesized Edition

2025-09-03

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1 Module 5: Scientific Investigations

Subject/Course: Investigating Science Stage 6 *Year Level:* 12 *Module:* 5

Source Docs: Investigating Science Stage 6 Syllabus (2017), National Literacy and Numeracy Learning Progressions, HSC Exams 2019-2024

1.1 Introduction

Module 5 provides a comprehensive framework for understanding and conducting complete scientific investigations. It addresses the fundamental questions of what initiates scientific inquiry, how to select appropriate methodologies, how to ensure investigation integrity through validity and reliability, and how to communicate findings through formal scientific reporting. This module synthesizes the observational and analytical skills developed in earlier modules into a structured approach for scientific research.

1.2 Inquiry Question 1 – What initiates an investigation?

1.2.1 Concepts

- Research factors that led scientists to investigate phenomena, including unexpected observations, theoretical predictions, technological advances, and practical needs
- Examine how initial observations transition into testable hypotheses through scientific reasoning
- Analyse the role of serendipity versus systematic research in scientific discovery
- Evaluate how deviations from traditional linear scientific methodology can lead to breakthrough discoveries

1.2.2 Definitions

- **Scientific investigation:** A systematic process of inquiry that begins with observation or a question, proceeds through hypothesis formation and testing, and concludes with evidence-based conclusions
- **Hypothesis:** A testable prediction about the relationship between variables, derived from observations or existing theory
- **Serendipity in science:** Accidental discoveries that occur when investigating unrelated phenomena, requiring a prepared mind to recognize their significance

1.2.3 Case Studies

Microwaves (Percy Spencer) Percy Spencer, working on radar technology during WWII, noticed a chocolate bar melting in his pocket while near a magnetron. This accidental observation led to systematic experiments with popcorn and eggs, ultimately resulting in the development of the microwave oven. This case demonstrates how unexpected observations, when pursued systematically, can lead to revolutionary technologies.

1.2.4 Practical Examples

- Temperature effects on enzyme activity initiated from observations of food spoilage
- Investigation of antibiotic resistance stemming from clinical treatment failures
- Research into ocean acidification triggered by coral bleaching observations

1.2.5 Links to Outcomes

- INS11-8: Identifies that the nature of science is tentative and dynamic
- INS12-12: Develops and evaluates the process of undertaking scientific investigations

1.2.6 Exam Alignment

- 2022 Q28: Factors initiating scientific investigation
- 2021 Q19: Role of observation in hypothesis formation

1.3 Inquiry Question 2 – What type of methodology best suits a scientific investigation?

1.3.1 Concepts

- Evaluate objectives and data types to determine appropriate investigative approaches
- Compare controlled experiments, observational studies, modelling, and field work methodologies
- Justify methodology selection based on hypothesis requirements, available resources, and ethical constraints
- Develop methods that balance ideal scientific design with practical limitations

1.3.2 Definitions

- **Methodology:** The systematic, theoretical analysis of methods applied to a field of study, encompassing the theoretical underpinning of chosen methods
- **Controlled experiment:** An investigation where all variables except the independent variable are kept constant to establish cause-effect relationships
- **Observational study:** Research where phenomena are observed without manipulation, used when experimentation is impossible or unethical
- **Field work:** Scientific investigation conducted in natural settings rather than controlled laboratory conditions

1.3.3 Case Studies

Clinical Trial Design The development of COVID-19 vaccines demonstrated multiple methodological approaches: laboratory research for vaccine development, controlled clinical trials for efficacy testing, and observational studies for real-world effectiveness monitoring.

1.3.4 Practical Examples

- Selecting titration for precise acid-base concentration determination
- Using transect sampling for ecological biodiversity assessment
- Employing spectroscopy for chemical composition analysis
- Choosing longitudinal studies for developmental research

1.3.5 Links to Outcomes

- INS12-2: Identifies and solves problems by generating testable questions and hypotheses
- INS12-12: Develops and evaluates the process of undertaking scientific investigations

1.3.6 Exam Alignment

- 2023 Q31: Methodology selection and justification
- 2020 Q26: Matching methodology to investigation type

1.4 Inquiry Question 3 – How is the integrity of a scientific investigation judged?

1.4.1 Concepts

- Evaluate investigation design for appropriate controls, sample size, and variable management
- Conduct investigations with careful attention to data collection protocols and error minimization
- Analyze primary data using appropriate statistical and graphical methods
- Assess validity through examination of methodology and variable control
- Evaluate reliability through repetition, replication, and consistency analysis
- Draw evidence-based conclusions and identify limitations

1.4.2 Definitions

- **Validity:** The extent to which an experiment measures what it claims to measure and directly tests the intended hypothesis. Key components include appropriate methodology, controlled variables, single independent variable manipulation, and relevant dependent variable measurement
- **Reliability:** The consistency and reproducibility of experimental results, achieved through sufficient repetition (minimum 3 trials), adequate sample size, and standardized procedures
- **Accuracy:** The closeness of measured values to the true or accepted value
- **Precision:** The closeness of repeated measurements to each other, regardless of their accuracy
- **Random error:** Unpredictable variations in measurements that affect precision but can be minimized through repetition
- **Systematic error:** Consistent bias in measurements that affects accuracy and cannot be eliminated through repetition

1.4.3 Case Studies

The Replication Crisis in Psychology Many landmark psychological studies have failed replication attempts, highlighting the importance of rigorous methodology, adequate sample sizes, and transparent reporting in ensuring scientific integrity.

1.4.4 Practical Examples

- **Valid experiment:** Testing plant growth with controlled light, water, temperature, and soil while varying only fertilizer concentration
- **Reliable experiment:** Measuring reaction rates with multiple trials, calculating mean and standard deviation
- **Invalid design:** Testing both temperature and pH effects simultaneously without proper controls

1.4.5 Links to Outcomes

- INS12-12: Develops and evaluates the process of undertaking scientific investigations
- INS11/12-4: Selects and processes appropriate qualitative and quantitative data

1.4.6 Exam Alignment

- 2021 Q22b: Validity in experimental design (GMO mice vs diet groups)
- 2020 Q25a: Reliability and sample size (self-injection example $n=1$)
- 2019 Q23: Distinguishing validity from reliability

1.5 Inquiry Question 4 – What is the structure of an investigative report?

1.5.1 Concepts

- Review peer-reviewed publications to identify scientific writing conventions
- Analyze the relationship between investigation components and report sections
- Understand the purpose and requirements of each report section
- Apply formal scientific language, appropriate terminology, and objective tone
- Present data using tables, graphs, and statistical analysis
- Construct logical arguments linking evidence to conclusions

1.5.2 Definitions

- **Scientific report structure:** Standardized format including Title, Abstract, Introduction, Aim, Hypothesis, Method, Results, Discussion, Conclusion, References
- **Objective writing:** Third-person, past tense narration avoiding personal pronouns and emotional language
- **Peer review:** Critical evaluation of scientific work by independent experts before publication
- **Citation:** Formal acknowledgment of sources using standardized referencing systems

1.5.3 Practical Examples

Report Section Requirements: - **Title:** Descriptive, concise, including key variables - **Abstract:** 150-250 word summary of entire investigation - **Introduction:** Background, literature review, rationale - **Aim:** Clear statement of investigation purpose - **Hypothesis:** Specific, testable prediction - **Method:** Detailed, reproducible procedure - **Results:** Objective data presentation without interpretation - **Discussion:** Analysis, comparison with literature, limitations - **Conclusion:** Summary addressing aim and hypothesis - **References:** Complete bibliographic information

1.5.4 Links to Outcomes

- INS11/12-7: Communicates scientific understanding using suitable language and terminology
- INS12-3: Conducts investigations to collect valid and reliable data

1.5.5 Exam Alignment

- 2023 Q30: Report conventions and scientific style
- 2022 Q34: Identifying report structure errors
- 2019 Q31: Purpose of discussion section

1.6 Connections to Previous Material

This module synthesizes skills from: - Module 1: Observation techniques and primary data collection - Module 2: Drawing inferences and using secondary sources - Module 3: Model construction and testing - Module 4: Understanding theories and laws that guide hypothesis formation

1.7 Outcomes & Assessment Alignment

Outcome Code	Strand	Outcome Statement	Assessment Links
INS12-12	Working Scientifically	Develops and evaluates the process of undertaking scientific investigations	2021 Q22b; 2020 Q25a
INS11/12-7	Communicating	Communicates scientific understanding using suitable language and terminology	2023 Q30
INS12-2	Planning	Identifies and solves problems by generating testable questions and hypotheses	2022 Q28
INS11/12-4	Processing Data	Selects and processes appropriate qualitative and quantitative data	2019 Q23

1.8 Conclusion / Preview

Module 5 establishes the foundation for rigorous scientific investigation. These skills are essential for: - Module 6: Evaluating how technology enhances and limits investigations - Module 7: Critically assessing scientific claims and evidence quality - Module 8: Understanding the broader societal context of scientific research

2 Module 6: Technologies

Subject/Course: Investigating Science Stage 6 *Year Level:* 12 *Module:* 6

Source Docs: Investigating Science Stage 6 Syllabus (2017), HSC Exams 2019-2024

2.1 Introduction

Module 6 explores the complex interrelationship between scientific advancement and technological development. It examines how measurement technologies determine data quality through their precision, accuracy, and inherent limitations. The module emphasizes the feedback loop where technological innovations enable new scientific discoveries, which subsequently drive further technological advancement. Students also consider safety protocols and the comparison between analog and digital technologies.

2.2 Inquiry Question 1 – How does technology enhance and/or limit scientific investigation?

2.2.1 Concepts

- Design investigations utilizing available technologies for measuring variables
- Evaluate technological limitations including range, resolution, and sensitivity
- Assess measurement devices for sources of random and systematic errors
- Compare analog versus digital technologies for accuracy and precision
- Apply safety protocols using Safety Data Sheets (SDS) and Work Health and Safety guidelines

2.2.2 Definitions

- **Measurement technology:** Instruments and devices used to quantify physical phenomena with defined precision and accuracy
- **Resolution:** The smallest change in a quantity that an instrument can detect
- **Sensitivity:** The ratio of instrument response to the measured quantity
- **Calibration:** The process of configuring an instrument to provide accurate measurements by comparison with standards
- **Analog technology:** Devices that represent data as continuous physical variables
- **Digital technology:** Devices that represent data as discrete numerical values
- **Safety Data Sheet (SDS):** Document containing information on potential hazards and safe use procedures for chemicals

2.2.3 Case Studies

Evolution of Temperature Measurement From mercury thermometers ($\pm 0.5^{\circ}\text{C}$) to digital sensors ($\pm 0.01^{\circ}\text{C}$) to infrared imaging, each technology has expanded investigation possibilities while introducing specific limitations and error sources.

2.2.4 Practical Examples

Effect of Temperature on Reaction Rate (Glow Stick Investigation) - Equipment: Glow sticks, water baths at different temperatures, light meter or camera - Observation: Higher temperatures increase reaction rate (brighter initial glow, shorter duration) - Limitations: Subjective brightness assessment without photometer, temperature control precision - Technology enhancement: Digital photometer provides quantitative data; thermal imaging shows heat distribution

2.2.5 Technology Comparison Table

Aspect	Analog	Digital
Reading	Continuous scale	Discrete display
Precision	Limited by scale divisions	High (multiple decimal places)
Response time	Instantaneous	Processing delay possible
Power requirement	Often none	Battery/power needed
Drift	Mechanical wear	Electronic drift
Data logging	Manual only	Automatic possible

2.2.6 Links to Outcomes

- INS12-3: Conducts investigations to collect valid and reliable primary and secondary data
- INS11/12-6: Solves problems using critical thinking and scientific processes

2.2.7 Exam Alignment

- 2024 Q35: Technology limitations in measurement
- 2023 Q32: Analog vs digital comparison
- 2021 Q24: Error sources in technological measurements

2.3 Inquiry Question 2 – How have developments in technology led to advances in scientific theories and laws?

2.3.1 Concepts

- Assess technology's impact on evidence accumulation for theories and models
- Evaluate how scientific advances drive technological innovation
- Analyze the cyclical relationship between scientific discovery and technological development
- Investigate Indigenous knowledge contributions to modern technologies
- Examine how new technologies reveal previously unobservable phenomena

2.3.2 Definitions

- **Science-technology feedback loop:** The reciprocal relationship where technological advances enable new scientific discoveries, which inspire further technological innovation
- **Enabling technology:** Instruments that make previously impossible observations or measurements feasible
- **Convergent technologies:** The synergistic combination of different technological fields to create new capabilities
- **Indigenous knowledge systems:** Traditional understanding of natural phenomena developed through systematic observation over generations

2.3.3 Case Studies

DNA Structure Discovery and Biotechnology Development 1. X-ray crystallography technology → Photo 51 (Rosalind Franklin) 2. DNA double helix model (Watson & Crick) 3. PCR technology development 4. CRISPR gene editing 5. Personalized medicine

This exemplifies the feedback loop: imaging technology revealed structure, understanding structure enabled manipulation technologies, which further refined our understanding.

2.3.4 Practical Examples

Technology-Enabled Discoveries: - Microscopy → Cell theory → Electron microscopy → Organelle discovery - Spectroscopy → Atomic structure → Laser technology → Quantum computing - Telescopes → Planetary motion → Space technology → Exoplanet detection - Particle accelerators → Subatomic particles → Medical imaging → Cancer treatment

2.3.5 Indigenous Knowledge Integration

- Plant medicines → Pharmaceutical development
- Fire management → Modern bushfire prevention
- Seasonal calendars → Climate science
- Water management → Sustainable agriculture

2.3.6 Links to Outcomes

- INS12-13: Describes and explains how science drives the development of technologies
- INS12-15: Evaluates implications of ethical, social, economic and political influences

2.3.7 Exam Alignment

- 2024 Q36: DNA structure and GM technologies (feedback loop)
- 2022 Q35: Technology enabling scientific discovery
- 2020 Q28: Indigenous knowledge and modern science

2.4 Connections to Previous Material

Building on Module 5's investigation design, Module 6 specifically addresses: - How technology choice affects validity and reliability - The relationship between measurement precision and conclusion strength - How technological limitations shape hypothesis formation

2.5 Outcomes & Assessment Alignment

Outcome Code	Strand	Outcome Statement	Assessment Links
INS12-13	Understanding	Describes how science drives technology development	2024 Q36
INS12-3	Conducting	Collects valid and reliable data	2023 Q32
INS11/12-6	Problem Solving	Uses critical thinking and scientific processes	2021 Q24

2.6 Conclusion / Preview

Understanding technology's role in science prepares students for Module 7's critical evaluation of evidence quality and Module 8's examination of how technological capabilities influence societal decisions about scientific research.

3 Module 7: Fact or Fallacy?

Subject/Course: Investigating Science Stage 6 *Year Level:* 12 *Module:* 7

Source Docs: Investigating Science Stage 6 Syllabus (2017), HSC Exams 2019-2024

3.1 Introduction

Module 7 develops critical thinking skills essential for navigating the modern information landscape. It addresses how to rigorously test claims, understand the factors that influence data interpretation, distinguish correlation from causation, and recognize pseudoscience. The module emphasizes the importance of peer review as science's quality control mechanism and examines how media representation affects public understanding of science.

3.2 Inquiry Question 1 – How can a claim be tested?

3.2.1 Concepts

- Design investigations to test specific claims using appropriate controls
- Evaluate the impact of sample selection and size on result validity
- Distinguish between evidence-based claims and emotional appeals
- Apply statistical analysis to determine claim validity
- Identify testable versus non-testable claims

3.2.2 Definitions

- **Claim:** An assertion about a phenomenon that can be tested through systematic investigation
- **Testable claim:** A statement that can be evaluated through empirical observation or experimentation
- **Sample size:** The number of observations in a study, affecting statistical power and generalizability
- **Sample bias:** Systematic error introduced when the sample doesn't represent the target population
- **Statistical significance:** The probability that observed differences are not due to chance alone
- **Effect size:** The magnitude of difference between groups, independent of sample size

3.2.3 Practical Examples

Testing Product Claims - Chocolate Chip Cookie Investigation - Claim: "Contains 40% chocolate chips by mass" - Method: 1. Weigh whole cookies (n 10) 2. Carefully separate and weigh chocolate chips 3. Calculate percentage: $(\text{chip mass}/\text{total mass}) \times 100\%$ 4. Statistical analysis: mean, standard deviation, confidence interval 5. Compare to claimed value using t-test

Sample Size Effects: - n=1: No statistical validity - n=3: Large uncertainty - n=10: Reasonable estimate - n=30: Statistical reliability - n=100: High confidence

3.2.4 Links to Outcomes

- INS12-14: Uses evidence-based analysis to support or refute hypotheses
- INS12-2: Identifies and solves problems by generating testable questions

3.2.5 Exam Alignment

- 2023 Q33: Testing commercial product claims
- 2021 Q25: Sample size and statistical validity
- 2019 Q24: Evidence-based vs emotional claims

3.3 Inquiry Question 2 – What factors affect data interpretation?

3.3.1 Concepts

- Justify the use of placebos, blinding, and control groups
- Evaluate societal and economic influences on data collection
- Understand confirmation bias and its effects
- Analyze the role of funding sources in research design
- Assess cultural factors in data interpretation

3.3.2 Definitions

- **Placebo:** An inactive treatment given to control groups to account for psychological effects
- **Placebo effect:** Improvement in condition due to belief in treatment rather than treatment itself
- **Single-blind trial:** Participants unaware of treatment allocation
- **Double-blind trial:** Neither participants nor researchers aware of treatment allocation
- **Control group:** Participants receiving no treatment or standard treatment for comparison
- **Confirmation bias:** Tendency to interpret data in ways that confirm preexisting beliefs
- **Funding bias:** Influence of financial sponsors on research design or interpretation

3.3.3 Case Studies

The Hawthorne Effect Workers at Western Electric’s Hawthorne plant showed productivity improvements regardless of environmental changes, revealing that observation itself affects behavior. This discovery revolutionized understanding of experimental design and the need for control groups.

3.3.4 Practical Examples

Clinical Trial Design Hierarchy: 1. **Open trial:** All parties aware (highest bias risk) 2. **Single-blind:** Participants unaware (moderate bias risk) 3. **Double-blind placebo-controlled:** Gold standard (minimal bias risk) 4. **Triple-blind:** Analysis also blinded (maximum objectivity)

3.3.5 Links to Outcomes

- INS11/12-5: Analyses and evaluates primary and secondary data
- INS12-6: Solves problems using critical thinking skills

3.3.6 Exam Alignment

- 2022 Q36: Double-blind trial necessity
- 2020 Q29: Placebo effect in medical trials
- 2019 Q26: Economic influences on research

3.4 Inquiry Question 3 – What type of evidence is needed for valid conclusions?

3.4.1 Concepts

- Evaluate the distinction between correlation and causation
- Identify confounding variables that create spurious correlations
- Understand criteria for establishing causation
- Assess evidence quality hierarchies
- Apply Bradford Hill criteria for causation

3.4.2 Definitions

- **Correlation:** A statistical relationship between two variables where they vary together
- **Causation:** A relationship where changes in one variable directly cause changes in another
- **Spurious correlation:** An apparent relationship between variables actually caused by a third factor
- **Confounding variable:** An unmeasured third variable affecting both variables under study
- **Temporal precedence:** The cause must occur before the effect
- **Dose-response relationship:** Stronger exposure leads to stronger effect
- **Biological plausibility:** A reasonable mechanism explains the causal relationship

3.4.3 Case Studies

Classic Correlation Causation Examples 1. **Ice cream sales and drowning deaths:** Both increase in summer (confounding variable: temperature) 2. **Shoe size and reading ability in children:** Both increase with age (confounding variable: development) 3. **Coffee consumption and heart disease:** Initial correlation disappeared when smoking was controlled

3.4.4 Bradford Hill Criteria for Causation

1. Strength of association
2. Consistency across studies
3. Specificity of relationship
4. Temporality
5. Biological gradient
6. Plausibility
7. Coherence
8. Experimental evidence
9. Analogy

3.4.5 Links to Outcomes

- INS12-14: Uses evidence-based analysis to support or refute hypotheses
- INS11/12-4: Selects and processes appropriate data

3.4.6 Exam Alignment

- 2019 Q25c: Correlation vs causation (whale populations and CO₂)
- 2021 Q26: Identifying confounding variables
- 2020 Q30: Evidence hierarchy in conclusions

3.5 Inquiry Question 4 – How does science reporting influence public understanding?

3.5.1 Concepts

- Compare peer-reviewed versus popular media reporting
- Analyze misuse of scientific terminology in media
- Evaluate how conflicts of interest affect reporting
- Understand the halo effect in science communication
- Identify characteristics of pseudoscience

3.5.2 Definitions

- **Pseudoscience:** Claims presented as scientific but lacking empirical support, falsifiability, or peer review
- **Halo effect:** Tendency for positive impression in one area to influence opinion in other areas
- **Cherry-picking:** Selecting only data that supports a predetermined conclusion
- **False balance:** Presenting opposing viewpoints as equally valid despite unequal evidence
- **Sensationalism:** Exaggerating findings to attract attention
- **P-hacking:** Manipulating data analysis to achieve statistical significance

3.5.3 Characteristics of Pseudoscience

1. **Unfalsifiable claims:** Cannot be proven wrong
2. **Reliance on anecdotes:** Individual stories over systematic data
3. **Cherry-picked evidence:** Ignoring contradictory data
4. **Absence of peer review:** Avoiding scientific scrutiny
5. **Static claims:** No evolution with new evidence
6. **Conspiracy theories:** Dismissing criticism as suppression
7. **Complex terminology:** Using jargon to appear scientific

3.5.4 Media Analysis Framework

- **Source credibility:** Author qualifications, institutional affiliation
- **Evidence quality:** Primary sources, sample size, methodology
- **Language analysis:** Certainty levels, qualifying statements
- **Conflict disclosure:** Funding sources, competing interests
- **Context provision:** Comparison with existing knowledge

3.5.5 Links to Outcomes

- INS11/12-6: Solves problems using critical thinking
- INS12-7: Communicates scientific understanding appropriately

3.5.6 Exam Alignment

- 2020 Q27a: Identifying pseudoscience (astrology example)
- 2022 Q37: Media misrepresentation of research
- 2019 Q27: Scientific terminology misuse

3.6 Inquiry Question 5 – Can peer review find ‘the truth’?

3.6.1 Concepts

- Analyze the peer review process and its limitations
- Evaluate the “publish or perish” culture
- Assess the replication crisis in science
- Understand predatory journals and their impact
- Examine post-publication review mechanisms

3.6.2 Definitions

- **Peer review:** Expert evaluation of scientific work before publication to ensure quality and validity
- **Reproducibility:** Ability to obtain consistent results using the same methods and data
- **Replicability:** Ability to obtain consistent results in new studies with same methods
- **Publish or perish:** Academic pressure to publish frequently for career advancement
- **Predatory journals:** Publications that charge fees without providing legitimate peer review
- **Retraction:** Formal withdrawal of published research due to errors or misconduct
- **Pre-registration:** Publishing study design before data collection to prevent p-hacking

3.6.3 The Peer Review Process

1. **Submission:** Author submits manuscript
2. **Editorial review:** Initial quality and scope assessment
3. **Reviewer selection:** 2-3 expert reviewers chosen
4. **Review:** Critical evaluation of methods, results, conclusions
5. **Decision:** Accept, reject, or revise
6. **Revision:** Authors address reviewer concerns
7. **Publication:** Final acceptance and publication

3.6.4 Peer Review Limitations

- **Time constraints:** Reviewers are unpaid volunteers
- **Expertise gaps:** Reviewers may lack specific knowledge
- **Bias potential:** Personal, institutional, or cultural biases
- **Innovation suppression:** Novel ideas may be rejected
- **Fraud detection:** Not designed to detect fabrication
- **Statistical review:** Often inadequate statistical expertise

3.6.5 Links to Outcomes

- INS11/12-5: Analyses and evaluates data and information
- INS12-15: Evaluates ethical and social influences on science

3.6.6 Exam Alignment

- 2019 Q29b: Peer review role (H. pylori discovery)
- 2021 Q27: Predatory journals impact
- 2023 Q34: Replication crisis implications

3.7 Connections to Previous Material

Module 7 synthesizes critical thinking skills with: - Module 5: Understanding what constitutes valid methodology - Module 6: Recognizing how technology can introduce bias - Previous modules: Applying observation and inference skills to evaluate claims

3.8 Outcomes & Assessment Alignment

Outcome Code	Strand	Outcome Statement	Assessment Links
INS12-14	Analysis	Uses evidence-based analysis to support/refute hypotheses	2019 Q25c
INS11/12-6	Problem Solving	Uses critical thinking and scientific processes	2020 Q27a
INS11/12-5	Evaluation	Analyses and evaluates data and information	2019 Q29b
INS12-5	Data Analysis	Processes primary and secondary data	2019 Q1, Q21a

3.9 Conclusion / Preview

Critical evaluation skills from Module 7 are essential for Module 8's examination of science's relationship with society, where students analyze how scientific evidence influences policy, ethics, and public decision-making.

4 Module 8: Science and Society

Subject/Course: Investigating Science Stage 6 *Year Level:* 12 *Module:* 8

Source Docs: Investigating Science Stage 6 Syllabus (2017), HSC Exams 2019-2024

4.1 Introduction

Module 8 examines the complex interface between scientific practice and societal context. It explores how scientific events shape public trust, the necessity of ethical and regulatory frameworks, and the influence of economic, social, and political forces on research. This module emphasizes that science does not occur in isolation but is deeply embedded in and influenced by the broader human context.

4.2 Inquiry Question 1 – How do science-related events affect society’s view of science?

4.2.1 Concepts

- Investigate how historical scientific events shaped public perception
- Analyze factors that build or erode public trust in science
- Evaluate the role of science communication in public understanding
- Assess how scientific failures and successes influence policy
- Understand the impact of scientific controversies on public confidence

4.2.2 Definitions

- **Public trust in science:** Society’s confidence in scientific institutions, methods, and findings
- **Science communication:** The practice of informing, educating, and raising awareness about science-related topics
- **Scientific controversy:** Disagreement within the scientific community or between science and society
- **Risk perception:** How the public understands and evaluates potential hazards
- **Science literacy:** The knowledge and understanding of scientific concepts and processes

4.2.3 Case Studies

Development of the Smallpox Vaccine (Edward Jenner, 1796) - Initial observation: Milkmaids with cowpox seemed immune to smallpox - **Controversial testing:** Inoculated 8-year-old James Phipps with cowpox, then exposed him to smallpox - **Public reaction:** Initial fear and resistance (“vaccination” derided as causing cow-like features) - **Long-term impact:** - Saved millions of lives - Led to global eradication (1980) - Established vaccination as cornerstone of public health - Created template for modern clinical trials and ethics

Other Transformative Events: 1. **Thalidomide tragedy (1960s):** Led to strict drug testing regulations 2. **Chernobyl (1986):** Shifted nuclear energy perception 3. **Human Genome Project (2003):** Raised genetic privacy concerns 4. **COVID-19 vaccine development (2020-21):** Demonstrated rapid science capability but also revealed trust issues

4.2.4 Links to Outcomes

- INS12-15: Evaluates implications of influences on science
- INS12-1: Develops and evaluates questions and hypotheses

4.2.5 Exam Alignment

- 2022 Q38: Public trust and vaccination
- 2020 Q31: Science communication during crises
- 2019 Q28: Historical events shaping science perception

4.3 Inquiry Question 2 – Why is scientific research regulated?

4.3.1 Concepts

- Investigate the historical development of research ethics
- Assess current ethical frameworks and their effectiveness
- Evaluate international codes of conduct and their enforcement
- Analyze the balance between research freedom and ethical constraints
- Understand special protections for vulnerable populations

4.3.2 Definitions

- **Research ethics:** Moral principles guiding scientific research involving humans, animals, or the environment
- **Informed consent:** Voluntary agreement to participate after understanding risks and benefits
- **Beneficence:** Obligation to maximize benefits and minimize harm
- **Justice:** Fair distribution of research benefits and burdens
- **Institutional Review Board (IRB):** Committee that reviews research proposals for ethical compliance
- **Declaration of Helsinki:** International ethical principles for medical research involving humans
- **The 3Rs:** Replacement, Reduction, Refinement in animal research

4.3.3 Ethical Frameworks

NHMRC National Statement (Australia, 2023) Core Principles: 1. **Research merit and integrity:** Justified, well-designed research 2. **Justice:** Fair participant selection and benefit distribution 3. **Beneficence:** Risk-benefit favorable ratio 4. **Respect:** Autonomy, consent, privacy, confidentiality

Historical Context - Why Regulation Exists: - **Nazi experiments (1940s):** Led to Nuremberg Code - **Tuskegee Study (1932-1972):** Untreated syphilis in African Americans - **Henrietta Lacks (1951):** Cells taken without consent - **Stanford Prison Experiment (1971):** Psychological harm to participants

4.3.4 Case Studies

Organ Transplantation Ethics - The challenge: Demand far exceeds supply (globally ~10% receive needed organs) - **Ethical frameworks developed:** - WHO Guiding Principles (1991, revised 2010) - National programs (e.g., Australia's DonateLife) - **Key principles:** - Voluntary consent (opt-in vs opt-out systems) - Prohibition of organ trafficking - Fair allocation algorithms - Living donor protection - **Ongoing issues:** - Black market trade (~10% of transplants globally) - Organ tourism - Xenotransplantation ethics - 3D bioprinting implications

4.3.5 Special Populations Protection

- **Children:** Cannot consent; require parental permission plus child assent
- **Pregnant women:** Dual consideration for mother and fetus
- **Prisoners:** Vulnerable to coercion
- **Indigenous peoples:** Collective consent, cultural sensitivity
- **Mentally impaired:** Capacity assessment required
- **Economically disadvantaged:** Protection from exploitation

4.3.6 Links to Outcomes

- INS12-15: Evaluates ethical implications on science
- INS11/12-1: Develops understanding of ethical frameworks

4.3.7 Exam Alignment

- 2020 Q25b: Ethics committee requirements (self-injection example)
- 2021 Q28: Vulnerable population protections
- 2019 Q30: International regulation effectiveness

4.4 Inquiry Question 3 – How do economic, social and political influences affect research?

4.4.1 Concepts

- Evaluate research funding allocation and priorities
- Analyze corporate influence on research agendas and outcomes
- Assess how political ideologies shape science policy
- Understand social movements' impact on research directions
- Examine the economics of publication and access to knowledge

4.4.2 Definitions

- **Conflict of interest:** Situation where personal or financial interests could compromise professional judgment
- **Research bias:** Systematic distortion in research design, conduct, or reporting
- **Lobbying:** Attempting to influence policy decisions, including research funding
- **Open science:** Making research methods, data, and findings freely accessible
- **Dual-use research:** Studies with both beneficial and potentially harmful applications
- **Valley of death:** Gap between basic research and commercial application

4.4.3 Economic Influences

Funding Sources and Their Effects: 1. **Government funding** (~60% in most developed nations) - Priorities reflect political agendas - Subject to budget constraints - Often basic research focused

2. **Industry funding** (~25-30%)

- Applied research emphasis
- Potential publication bias
- Intellectual property concerns

3. **Philanthropic funding** (~5-10%)

- Can support neglected areas
- May reflect donor interests
- Often disease-specific

The “Publish or Perish” Economy: - Average academic publishes 2-3 papers/year - Career advancement tied to publication metrics - Journal publishing: \$10 billion industry - Article processing charges: \$1,500-5,000 - Subscription costs: Major universities pay \$1-2 million/year

4.4.4 Political and Social Influences

Examples of Political Impact: 1. **Climate science:** Funding and acceptance varies with political leadership 2. **Stem cell research:** Religious and ethical views shape policy 3. **Defense research:** Military priorities drive technology development 4. **Space exploration:** National prestige influences funding

Franklin River Dam Case (Tasmania, 1983) - Scientific evidence: Unique ecosystem, Aboriginal heritage sites - Political conflict: State development vs federal environment protection - High Court decision: Commonwealth power to stop dam - Impact: Established precedent for science in environmental decisions

4.4.5 Conflict of Interest Examples

Type	Example	Impact
Financial	Tobacco industry funding respiratory research	Delayed recognition of smoking harms
Professional	Peer reviewers competing for same grants	Biased manuscript evaluations
Institutional	Universities dependent on donor funding	Pressure to produce favorable results
Political	Government scientists during policy debates	Censorship of contradictory findings
Personal	Researchers with patents on study drugs	Overstatement of benefits

4.4.6 Corporate Influence Case Studies

1. Pharmaceutical industry:

- Funds 60% of drug trials
- Positive results 4x more likely to be published
- Ghost-writing of academic papers

2. Food industry:

- Sugar industry suppressed heart disease links (1960s)
- Influenced dietary guidelines

3. Fossil fuel industry:

- Climate denial funding: \$2 billion (2000-2010)
- Created doubt despite scientific consensus

4.4.7 Links to Outcomes

- INS12-15: Evaluates economic and political influences
- INS12-14: Uses evidence-based analysis
- INS11/12-2: Proposes solutions to scientific problems

4.4.8 Exam Alignment

- 2022 Q33b: Corporate funding conflicts
- 2020 Q32c: Climate communication conflicts
- 2019 Q30b: Franklin Dam - science and politics

4.5 Connections to Previous Material

Module 8 integrates all previous modules: - Module 5: How ethical considerations shape methodology
- Module 6: How economic factors influence technology access - Module 7: How societal factors affect evidence interpretation - All modules: The human context of scientific practice

4.6 Outcomes & Assessment Alignment

Outcome Code	Strand	Outcome Statement	Assessment Links
INS12-15	Evaluation	Evaluates ethical, social, economic, political influences	2019 Q30b; 2020 Q25b, Q32c; 2022 Q33b
INS12-14	Analysis	Uses evidence-based analysis	Multiple questions
INS11/12-1	Understanding	Develops understanding of science nature	2022 Q38

4.7 Conclusion

Module 8 completes the Investigating Science course by placing scientific practice within its full societal context. Students should now understand that: - Science is a human endeavor subject to human influences - Ethical frameworks protect both research subjects and scientific integrity - Economic and political forces shape what research is conducted - Public trust in science requires transparency, ethics, and effective communication - Scientific evidence must compete with other factors in policy decisions

This understanding prepares students to be informed citizens who can critically evaluate scientific claims and participate in science-related societal decisions.

5 Comprehensive Definitions Glossary

5.1 Core Investigation Concepts

Accuracy: The degree to which measured values correspond to the true or accepted value of the quantity being measured. Accuracy is affected by systematic errors and calibration.

Calibration: The process of configuring an instrument to provide accurate measurements by comparison with known standards, essential for maintaining measurement accuracy.

Causation: A relationship where changes in one variable directly produce changes in another variable, established through controlled experimentation and meeting specific criteria such as temporal precedence and dose-response relationships.

Claim: An assertion about a phenomenon that can be tested through systematic investigation, forming the basis for hypothesis development.

Confounding variable: An unmeasured third variable that affects both the independent and dependent variables, potentially creating spurious correlations.

Control group: The group in an experiment that receives no treatment or standard treatment, providing a baseline for comparison with experimental groups.

Controlled experiment: An investigation where all variables except the independent variable are kept constant to establish cause-effect relationships.

Correlation: A statistical relationship between two variables where they vary together, which does not necessarily imply that one causes the other.

Hypothesis: A testable prediction about the relationship between variables, derived from observations or existing theory and stated in a form that can be supported or refuted through experimentation.

Independent variable: The factor that is deliberately manipulated in an experiment to observe its effect on the dependent variable.

Dependent variable: The factor that is measured or observed in an experiment to determine the effect of the independent variable.

Methodology: The systematic, theoretical analysis of methods applied to a field of study, encompassing the rationale and philosophical assumptions underlying chosen research methods.

Observational study: Research where phenomena are observed and recorded without manipulation of variables, used when experimentation is impossible, impractical, or unethical.

Placebo: An inactive treatment given to control groups to account for psychological effects of receiving treatment, essential for determining true treatment efficacy.

Placebo effect: Improvement in condition due to belief in treatment effectiveness rather than the treatment itself, demonstrating the power of psychological factors in perceived outcomes.

Precision: The closeness of repeated measurements to each other, regardless of their accuracy, affected by random errors and measurement resolution.

Primary data: Information collected directly by the researcher through experimentation or observation for a specific investigation.

Random error: Unpredictable variations in measurements that affect precision but can be minimized through repetition and statistical analysis.

Reliability: The consistency and reproducibility of experimental results, achieved through sufficient repetition (minimum 3 trials), adequate sample size, and standardized procedures.

Resolution: The smallest change in a quantity that an instrument can detect, determining the precision limits of measurements.

Sample bias: Systematic error introduced when the sample doesn't represent the target population, affecting the generalizability of results.

Sample size: The number of observations or replicates in a study, affecting statistical power, reliability, and the generalizability of conclusions.

Scientific investigation: A systematic process of inquiry that begins with observation or a question, proceeds through hypothesis formation and testing, and concludes with evidence-based conclusions.

Secondary data: Information that has been collected by others and is used to support or validate primary findings.

Sensitivity: The ratio of instrument response to the quantity being measured, determining the minimum detectable change.

Statistical significance: The probability that observed differences between groups are not due to chance alone, typically set at $p < 0.05$.

Systematic error: Consistent bias in measurements that affects accuracy and cannot be eliminated through repetition, requiring calibration or methodological changes to correct.

Validity: The extent to which an experiment measures what it claims to measure and directly tests the intended hypothesis through appropriate methodology, controlled variables, and relevant measurements.

Variable: Any factor that can change or be changed in an experiment, classified as independent (manipulated), dependent (measured), or controlled (kept constant).

5.2 Technology and Measurement

Analog technology: Devices that represent data as continuous physical variables, offering immediate response but limited precision.

Convergent technologies: The synergistic combination of different technological fields to create new capabilities exceeding the sum of their parts.

Digital technology: Devices that represent data as discrete numerical values, offering high precision and data logging capabilities but requiring power and potentially having processing delays.

Enabling technology: Instruments and techniques that make previously impossible observations or measurements feasible, opening new areas of investigation.

Measurement technology: Instruments and devices used to quantify physical phenomena with defined precision, accuracy, range, and resolution.

Safety Data Sheet (SDS): Comprehensive document containing information on potential hazards, safe handling procedures, and emergency responses for chemicals.

Science-technology feedback loop: The reciprocal relationship where technological advances enable new scientific discoveries, which in turn inspire further technological innovation.

5.3 Critical Thinking and Evidence

Cherry-picking: The practice of selecting only data that supports a predetermined conclusion while ignoring contradictory evidence.

Confirmation bias: The tendency to interpret data in ways that confirm preexisting beliefs, a major source of error in research interpretation.

Dose-response relationship: The principle that stronger exposure to a causal agent produces stronger effects, one criterion for establishing causation.

Effect size: The magnitude of difference between groups, independent of sample size, indicating practical significance.

Evidence hierarchy: The ranking of evidence types by reliability, from systematic reviews (highest) to expert opinion (lowest).

False balance: Media practice of presenting opposing viewpoints as equally valid despite unequal supporting evidence.

Falsifiability: The ability of a hypothesis or theory to be proven wrong through testing, a key criterion for scientific claims.

Halo effect: The tendency for a positive impression in one area to influence opinion in other areas, affecting objectivity.

P-hacking: Manipulating data analysis methods to achieve statistical significance, undermining research integrity.

Peer review: The evaluation of scientific work by independent experts in the same field to ensure quality and validity before publication.

Pseudoscience: Claims, beliefs, or practices presented as scientific but lacking empirical support, falsifiability, or adherence to the scientific method.

Publication bias: The tendency for positive results to be published more frequently than negative results, skewing the scientific literature.

Replicability: The ability to obtain consistent results in new studies using the same methods, essential for confirming findings.

Reproducibility: The ability to obtain consistent results using the original data and methods, a minimum standard for scientific validity.

Spurious correlation: An apparent relationship between variables actually caused by a third factor rather than direct causation.

Temporal precedence: The requirement that a cause must occur before its effect, one criterion for establishing causation.

5.4 Ethics and Society

Beneficence: The ethical principle requiring researchers to maximize benefits and minimize harm to research participants.

Conflict of interest: A situation where personal, financial, or professional interests could compromise scientific objectivity or professional judgment.

Declaration of Helsinki: International ethical principles for medical research involving human subjects, establishing global standards.

Dual-use research: Scientific studies with both beneficial applications and potential for harm, requiring special ethical consideration.

Ethical research principles: Core principles governing research: informed consent, beneficence, justice, and respect for persons.

Funding bias: The influence of research sponsors on study design, conduct, or interpretation, potentially compromising objectivity.

Informed consent: Voluntary agreement to participate in research after understanding all risks, benefits, and procedures.

Institutional Review Board (IRB): Committee that reviews research proposals for ethical compliance and participant protection.

Justice: The ethical principle requiring fair selection of research participants and equitable distribution of research benefits and burdens.

Open science: The practice of making research methods, data, and findings freely accessible to all, promoting transparency and collaboration.

Predatory journals: Publications that charge fees without providing legitimate peer review, undermining scientific quality control.

Public trust in science: Society's confidence in scientific institutions, methods, and findings, essential for science's social license.

Research bias: Systematic distortion in research design, conduct, analysis, or reporting that affects conclusions.

Research ethics: The moral principles and guidelines governing the conduct of scientific research.

Retraction: The formal withdrawal of published research due to errors, misconduct, or ethical violations.

Science communication: The practice of informing, educating, and engaging the public about science-related topics.

Science literacy: Public understanding of scientific concepts, methods, and the nature of science necessary for informed citizenship.

Social and political influence: The impact of societal values, political ideologies, and cultural factors on scientific research priorities and interpretation.

The 3Rs: Ethical principles for animal research: Replacement (use alternatives), Reduction (minimize numbers), Refinement (minimize suffering).

6 Study Guide Summary

6.1 Module Progression and Connections

Year 11 Foundation (Modules 1-4): - Module 1: Observation skills - Module 2: Inference and interpretation - Module 3: Scientific models - Module 4: Theories and laws

Year 12 Application (Modules 5-8): - Module 5: Complete investigation cycle - Module 6: Technology's role - Module 7: Critical evaluation - Module 8: Societal context

6.2 Key Skills Development

1. **Investigation Design:** Planning, conducting, and evaluating experiments with attention to validity and reliability
2. **Data Analysis:** Processing quantitative and qualitative data using appropriate statistical methods
3. **Critical Thinking:** Evaluating claims, recognizing bias, distinguishing correlation from causation
4. **Scientific Communication:** Writing formal reports, using appropriate terminology, presenting data effectively
5. **Ethical Reasoning:** Understanding research ethics, identifying conflicts of interest, evaluating societal impacts
6. **Technology Application:** Selecting appropriate tools, understanding limitations, recognizing the science-technology feedback loop

6.3 Essential Understanding for HSC Success

- Master the distinction between validity (correct method) and reliability (consistent results)
- Understand that correlation does not imply causation without additional evidence
- Recognize the characteristics of pseudoscience versus genuine scientific claims
- Appreciate the role of peer review as quality control, not truth determination
- Understand how economic, political, and social factors influence scientific practice
- Apply ethical principles to evaluate research proposals and practices
- Use the science-technology feedback loop to explain historical and contemporary developments

6.4 Exam Preparation Focus Areas

1. **Experimental Design:** Be able to identify flaws in methodology and suggest improvements
2. **Data Interpretation:** Analyze graphs, tables, and statistical results critically
3. **Scientific Communication:** Know report structure and appropriate language conventions
4. **Evidence Evaluation:** Assess the quality and relevance of different evidence types
5. **Ethical Analysis:** Apply frameworks to research scenarios
6. **Technology Assessment:** Evaluate how tools affect investigation outcomes
7. **Societal Connections:** Link scientific developments to broader social impacts